ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 6 · Simple Exponential Smoothing

Professor: Angel Mora Segura

01 October 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA ✓
3	23 Sep	Lags, ACF, PACF ✓
4	24 Sep	Decomposition, STL, Box-Cox · Checkpoint 1 ✓
5	30 Sep	Benchmarks: Naive & Seasonal Naive ✓
6	01 Oct	Simple exponential smoothing **TODAY**
7	07 Oct	Holt, damped trend, Holt-Winters
8	08 Oct	ETS taxonomy + model selection
9	15 Oct	**Midterm (intermediate test) — 20% · Checkpoint 2**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier
17	25 Nov	Integrated Forecasting Workshop · Checkpoint 3
18	02 Dec	**Group presentations — 30% · double session, part 1**
19	02 Dec	**Group presentations — 30% · double session, part 2**
20	11 Dec	**Final exam in R — 40%**

TODAY'S SESSION

Weighting the Past

A weighted average where the weights decay with age, one parameter that controls how fast, and the data choosing that parameter for you.

FPP3 · Sections 8.1 – 8.2

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Explain simple exponential smoothing as a **weighted average with exponentially decaying weights**, and place it between the naive and mean methods.

2. Interpret the smoothing parameter **α**: what a high and a low value mean for how fast the forecast reacts.

3. Fit SES in R, let the data **estimate α**, and read the fitted level and its prediction intervals.

4. Recognise when SES is appropriate — and the **trend and seasonality** it cannot handle.

5. **Test SES against your benchmark** and record whether it earns a place in your recommendation.

BLOCK 1 · 30 MINUTES

Concept + R Application

One parameter, one idea, and the most widely used forecasting method in industry.

IN CLASS: THE CORE IDEAS · ANNEX + QUARTO: EQUATIONS, R DETAILS AND THE REST

Between the Two Extremes

The naive and mean methods make opposite assumptions about the past. Simple exponential smoothing sits between them, and a single parameter decides where.

NAIVE

Only the last value counts

All the weight on one observation. Reacts instantly to every change — including every bit of noise.

SIMPLE EXPONENTIAL SMOOTHING

Recent counts more, old counts less

Every observation contributes, with weights that shrink the further back it is. Reacts to real changes, dampens the noise.

MEAN

Every value counts equally

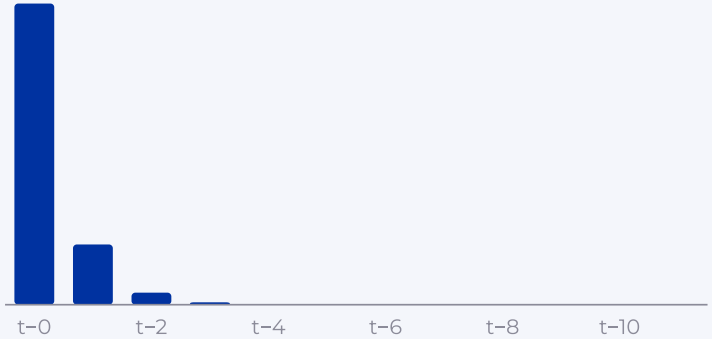
A value from three years ago weighs as much as yesterday's. Stable, but slow to notice anything has changed.

The smoothing parameter α slides between the two: α close to 1 behaves like naive, α close to 0 like the mean.

Weights That Decay With Age

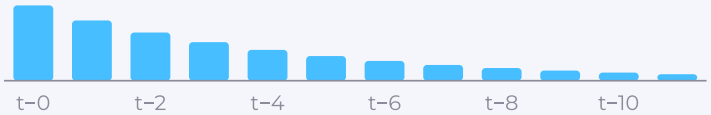
Each observation's weight is α times the weight of the one after it, scaled by (1 − α). The weights always add up to one, so the forecast is a genuine average.

α = 0.8 · reacts fast



80% of the weight on the latest observation; almost nothing beyond three periods back.

α = 0.2 · reacts slowly

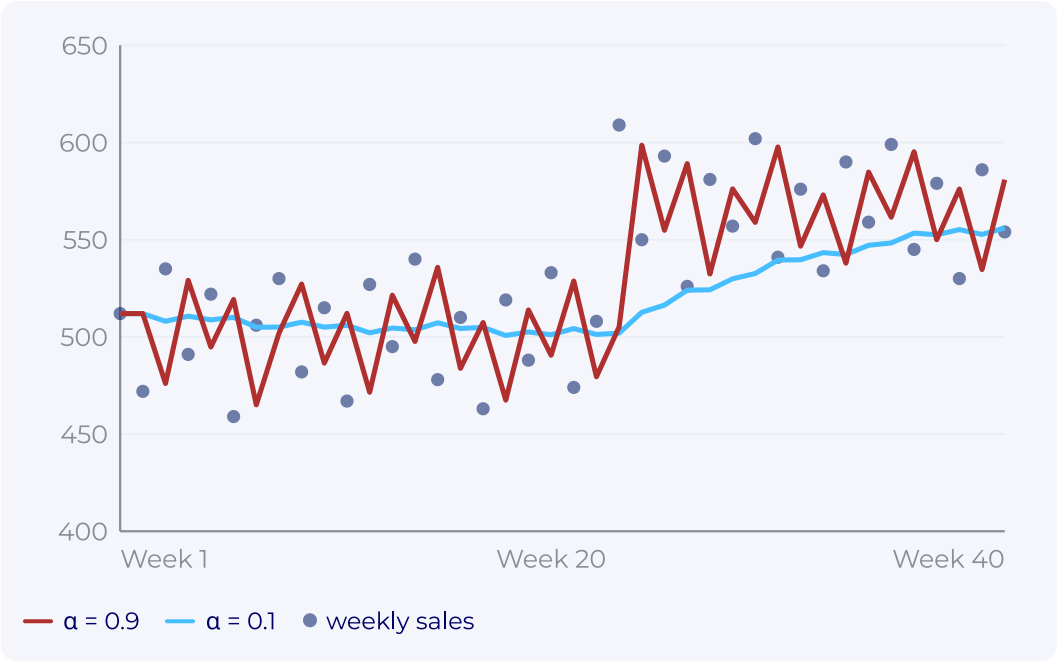


The weight is spread across many past periods, so one unusual value barely moves the forecast.

The name comes from the shape: the weights fall away exponentially, not in a straight line.

EXAMPLE 40 WEEKS OF SALES AT A STABLE STORE, WITH A LEVEL SHIFT AT WEEK 23

What α Does to the Fitted Level



The trade-off in one picture.

- **α = 0.9** chases every week. It catches the shift immediately, but it also copies the noise into the forecast.
- **α = 0.1** ignores the noise, but takes many weeks to notice the level has moved up.
- The data-chosen value here is **α ≈ 0.25**: the one that minimises the one-step errors on the training data.

EXAMPLE TRAINED ON WEEKS 1–32, FORECAST WEEKS 33–40

A Flat Line and a Widening Fan



Every horizon gets the same point forecast.

- SES has no idea of direction or season, so its best guess for next week and for eight weeks ahead is the same: the current level.
- The **interval** is what changes. It widens with the horizon, faster when α is large, because each step compounds the uncertainty.
- For a stable series this is exactly right. For a trending or seasonal one it is systematically wrong — next slide.

When Simple Exponential Smoothing Is the Wrong Tool

SES assumes a level that wanders, with no trend and no season. Use the diagnostics from Block 1 to decide before fitting it.

APPROPRIATE

No trend, no seasonality

Stable demand, a noisy level, a series whose ACF decays quickly. SES is often the best method available here.

TREND

The forecast lags behind

SES keeps forecasting the last level while the series keeps rising. Errors grow with the horizon. Holt's method adds a trend — Session 7.

SEASONALITY

The peaks are flattened

A flat line cannot reproduce a summer peak. Seasonal naive will usually beat it. Holt-Winters adds the season — Session 7.

WORKAROUND

Smooth the adjusted series

Remove the seasonality with STL, smooth what is left with SES, and add the season back. Next slide.

Using SES on a Seasonal Series

Most of your cases are seasonal. You can still use SES today by combining it with the decomposition from Session 4 — and it is often a strong competitor to seasonal naive.

```
fit <- train |>
  model(
    snaive  = SNAIVE(units),
    stl_ses = decomposition_model(
      STL(log(units)),
      ETS(season_adjust ~ error("A") +
          trend("N") + season("N"))
    )
  )
fit |> forecast(h = 12) |>
  accuracy(sales)
```

- **STL** removes the seasonal pattern. **SES** smooths the level of what remains. The seasonal component is added back with seasonal naive automatically.
- Compared with seasonal naive, it replaces “same month last year” with “same seasonal shape, current level” — useful when the level has drifted.
- It still cannot extrapolate a trend: the adjusted series is forecast flat. If your trend panel in Session 4 was clearly rising, expect it to under-forecast.

For the lab: this is the SES variant most teams should test against their benchmark today.

EXAMPLE TEST SET · WEEKS 33–40

Does It Earn Its Place?

Method	RMSE	MAE	MASE
SES	**25.5**	**21.5**	**0.45**
Naive	40.7	34.8	0.73
Mean	51.3	46.0	0.97

On a stable series with a level shift, SES wins clearly.

- Naive reacted to the shift but copied the noise of week 32 forward. Mean averaged over both levels and forecast too low. SES landed in between, which was where the truth was.
- The MAE falls from 34.8 to 21.5 units — a 38% improvement on the best benchmark.
- On a strongly seasonal series the same table would favour seasonal naive. **The table decides, not the name of the method.**

BLOCK 2 · 30 MINUTES

Business Case Lab

Add SES to your accuracy table and find out whether it beats the benchmark you fixed last session.

WHAT TO EXPECT IN EACH CASE

SES and Your Case

CASE 1 · RETAIL

Happy Cow Ice Cream

Daily cups have a strong weekly cycle. Plain SES will lose to seasonal naive; test SES on the STL-adjusted series with m = 7.

Watch for: the temperature effect ends up in the adjusted series. A high α may be SES chasing the weather.

CASE 2 · AIRLINES

Balearic Airlines

Same structure as Happy Cow: weekly seasonality, so use the STL-adjusted variant.

Watch for: the protection decision needs intervals, not the flat line. Look at how wide the 80% interval is at 9–15 days.

CASE 3 · MANUFACTURING

Clean Air Limited

Try both: SES on the total, and SES on the seasonally adjusted series. With 24 training months, the estimated α will be imprecise.

Watch for: the structural break. SES adapts to a new level, which may be exactly what a post-COVID series needs.

CASE 4 · E-COMMERCE

Wayfair

SES is the classic method for slow-moving items. Fit it to the four Decor SKUs and the category total, and compare with the benchmarks by MASE.

Watch for: SES on intermittent demand forecasts a small positive number every month. That is an average rate, not a prediction of any single month.

BLOCK 3 · 20 MINUTES

Case Clinic

Three or four teams: did SES beat your benchmark, and what does your α say about the business?

SESSION 6 RECAP

What to Take Away

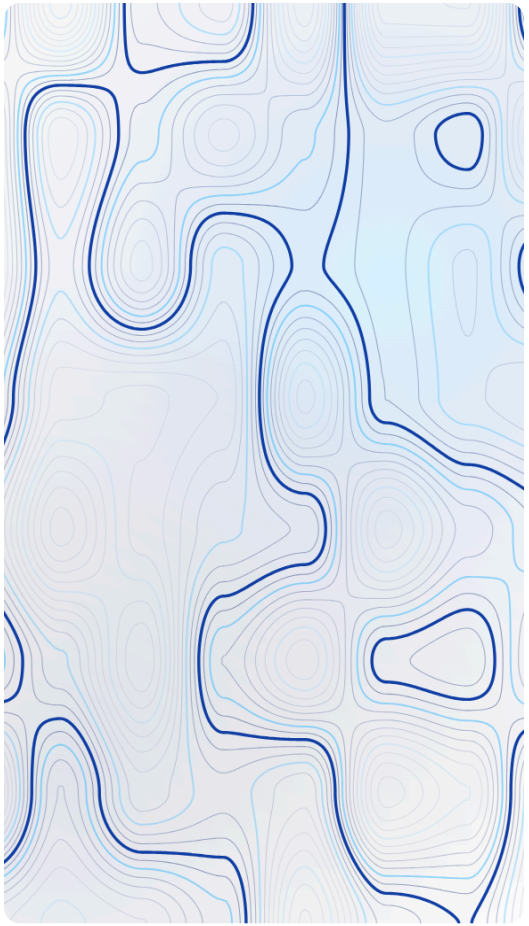
A weighted average of the past. Weights decay exponentially with age, sitting between the naive and mean methods.

α is estimated, and it means something. High: the series moves and recent values dominate. Low: mostly noise around a stable level.

The forecast is flat, the interval is not. Every horizon gets the current level; the uncertainty widens with each step.

No trend, no season — unless you deseasonalise. For seasonal cases, smooth the STL-adjusted series.

Next session: Holt adds a trend, Holt-Winters adds a season — built on exactly the same level equation you saw today.



END OF SESSION 06

TIME SERIES ANALYSIS

Simple Exponential Smoothing

NEXT · SESSION 7 · 07 OCTOBER

Holt and Holt-Winters

Smoothing learns to follow a trend and a seasonal pattern — the two things today's method cannot do.

The annex after this slide holds the reference material, including the R cheat sheet on slide 28.

Angel Mora Segura · IE University

01 October 2026

ANNEX · REFERENCE MATERIAL

For Study, Not for Class

Equations, R details, the cheat sheet and the common mistakes. Everything here is also developed in the session’s Quarto document.

The Method in Two Equivalent Forms

The same method can be written as a weighted average or as a level that updates every period. The second form is the one that generalises to trend and seasonality in the next two sessions.

Weighted-average form

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1-\alpha)y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \dots$$

Makes the decaying weights explicit. Good for understanding; awkward to compute.

Component form

Forecast: $\hat{y}_{t+h|t} = l_t$

Level: $l_t = \alpha y_t + (1-\alpha) l_{t-1}$

The level l is updated each period as a blend of the new observation and the previous level. Holt and Holt-Winters add a trend and a season to this same equation.

Every forecast from SES is the current level, for every horizon. That is why the forecast is a flat line.

Letting the Data Choose α

You do not pick α by eye. R tries every value between 0 and 1, produces the one-step fitted values for each, and keeps the one with the smallest sum of squared errors — together with the starting level l_0.

WHAT IS MINIMISED

One-step errors on the training data

For each candidate α, how far each fitted value was from the observation that followed. The best α is the one that tracked the past most closely.

WHAT IT TELLS YOU

How noisy the series is

A small estimated α says the series is mostly noise around a stable level. A large one says it moves, and recent values carry most of the information.

IN THE EXAMPLE

α ≈ 0.25

Low enough to ignore the week-to-week noise, high enough to catch the level shift within a few weeks.

Estimated α is a **finding about the business**, not only a parameter. Write it in the log with one line of interpretation.

R APPLICATION

Fitting SES in the Same Pipeline

```
fit <- train |>
  model(
    ses   = ETS(sales ~ error("A") +
                trend("N") + season("N")),
    naive = NAIVE(sales),
    mean  = MEAN(sales)
  )
fit |> select(ses) |> report()
#> Smoothing parameters:
#>   alpha = 0.25
#> Initial states:
#>   l[0] = 503.2
```

Reading the code

- SES is written as an **ETS** model: additive error, no trend, no season. That notation is the whole of Session 8; for now, read it as “simple exponential smoothing”.
- `report()` prints the estimated α and the starting level.
- `components()` gives the level at every period — plot it to see how the method moved.
- The benchmarks sit in the same `model()` call, so one `accuracy()` compares them all.

EVERYTHING FROM TODAY, IN ONE PLACE

Session 6 R Cheat Sheet

```r
# 1. fit SES next to the benchmarks
fit <- train |> model(
  ses = ETS(y ~ error("A") +
            trend("N") + season("N")),
  naive  = NAIVE(y),
  snaive = SNAIVE(y))
# 2. the estimated parameters
fit |> select(ses) |> report()
fit |> select(ses) |> tidy()
# 3. the fitted level over time
fit |> select(ses) |> components() |>
  autoplot()
```

```r
# 4. SES on a seasonal series
fit2 <- train |> model(
  stl_ses = decomposition_model(
    STL(log(y)),
    ETS(season_adjust ~ error("A") +
        trend("N") + season("N"))))
# 5. forecast, intervals, accuracy
fc <- fit |> forecast(h = 12)
fc |> autoplot(df, level = 80)
fc |> accuracy(df) |> arrange(MASE)
# 6. residuals
fit |> select(ses) |> gg_tsresiduals()
```

Four Common Misreadings

SES is simple enough that its mistakes are easy to make and easy to miss.

Fitting SES to a seasonal series directly	The flat forecast wipes out the peaks, and seasonal naive wins. Deseasonalise first, or wait for Holt-Winters.
Reading a high α as a good model	A high α only says recent values dominate. It can equally mean the series is close to a random walk, where naive does just as well.
Ignoring the intervals because the line is flat	The point forecast does not move, but the uncertainty grows with every step. Report the interval for the horizon the decision needs.
Declaring victory on training accuracy	SES chooses α to fit the training data as closely as possible, so its in-sample errors always look good. Only the test set counts.